SOP-EMBEDDED PROCESS

[CONFIDENTIAL]
[UTTHUNGA]
[IndiQube, South Summit, South End Rd, Vijayarangam Layout, Bengaluru, Karnataka 560004]

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CHANGE HISTORY

This section describes the details of changes that have resulted in the current document.

SL. No.	Document Version No.	Change Description	Reason For change	Date Prepared By	Date Reviewed By	Date Approved By
1	0.1	Draft	Implementation of the Integrated Management System	Siva Prasad 28-Dec-23		
2	Issue B Version 1.1	Review comments			Anitha N 07-Mar-24	
3	Issue B Version 1.2	Approved				Santosh V 08-Mar-24
4	Issue B Version 1.3	Document formatting done as per document guidelines	Updating Document as per Document guidelines	Shareef Shaik 22-Dec-24	Anitha N 31-Dec-24	Santosh V 31-Dec-24
5	Issue C Version 1.0	Review of document	New version update of IMS	Anitha N 01-08-2025	Santosh V 08-Aug-25	Sanmangala MS 14-08-2025

1 Purpose

This document establishes standardized processes and procedures for executing various tasks and activities throughout the embedded project lifecycle.

2 Scope

The scope of this document is to ensure consistency, quality, and efficiency throughout the embedded project development process, providing a comprehensive guide for engineers and stakeholders involved in the project.

3 Abbreviations

a. CR: Change Request

b. BOM: Bill of Materials

c. DRC: Design Rule Check

d. PCB: Printed Circuit Board

e. SI: Signal Integrity

f. PI: Power Integrity

g. MTBF: Mean Time Between Failures

h. FMEA: Failure Modes and Effects Analysis

i. DXF: Drawing Exchange Format

j. DVT: Design Verification Testing

k. EMI: Electromagnetic Interference

l. EMC: Electromagnetic Compatibility

m. LoRa: Long Range

n. ECR: Engineering Change Request

o. ECO: Engineering Change Order

p. CCB: Change Control Board

q. EMS: Electronic Manufacturing Services

r. OEMs: Original Equipment Manufacturers

s. API: Application Programming Interface

t. POC: Proof of Concept

u. DAR: Design Analysis Report

4 Terms and Definitions

a. Bill of Materials (BOM): A comprehensive list of all materials, components, and assemblies required to produce an embedded system, including part numbers and quantities.

b. Design Rule Check (DRC): The process of verifying that the design layout adheres to specific rules and constraints to ensure manufacturability and functionality.

c. Printed Circuit Board (PCB): A board that mechanically supports and electrically connects electronic components using conductive pathways etched from copper sheets laminated onto a non-conductive substrate.

d. Silicon (Si): The material commonly used in semiconductor manufacturing, forming the basis of integrated circuits and other electronic components.

e. Power Integrity (PI): The study and management of power distribution and consumption in electronic systems to ensure stable and reliable power delivery.

f. Mean Time Between Failures (MTBF): The average time between failures of a system or component, indicating its reliability.

g. Failure Modes and Effects Analysis (FMEA): A systematic method for evaluating potential failure modes of a system and their impact on system performance.

h. Drawing Exchange Format (DXF): A file format used for exchanging CAD (Computer-Aided Design) drawings between different systems.

i. Electronic Manufacturing Services: Range of services provided by companies that specialize in manufacturing electronic components and products for original equipment manufacturers (OEMs).

5 Responsibilities

a. Project Manager: shall oversee the embedded project development lifecycle. PM shall involve close coordination with various stakeholders, including clients, suppliers, and team members. The Project Manager is responsible for managing project timelines, budgets, and resource allocation.

b. Hardware Engineer: Responsible for designing high-level architecture and schematic diagrams, ensuring a comprehensive understanding of the project's hardware needs. The Hardware Engineer actively engages in the selection of components, performing detailed component comparisons, and creating the Bill of Materials (BOM)., they collaborate closely on PCB design, layout, and routing, contributing their expertise to the development of a robust and effective hardware system for the embedded project.

c. Firmware Engineer: Primary responsibility is to develop firmware for embedded systems, ensuring optimal functionality and performance. This involves implementing board support packages and actively participating in the board bring-up process. The Firmware Engineer collaborates closely with hardware

engineers, contributing to activities such as pin configuration and application porting to ensure seamless integration. Additionally, a key focus is placed on adhering to secure coding practices to enhance the overall security of the embedded system.

d. Mechanical Engineer: Is responsible for collaborating with other engineers, such as hardware and firmware engineers, to ensure that mechanical designs and optimize enclosures align considering factors like size, shape, material, and thermal management and integrate mechanical components seamlessly with electronic and firmware components, ensuring that the overall system functions as intended.

e. Test Engineer: Responsible for comprehensive test planning and execution for both hardware and firmware components. The Test Engineer conducts meticulous visual and physical inspections of Printed Circuit Board Assemblies (PCBA) to ensure quality standards. Active involvement in board bring-up, functional testing, and performance testing is a key responsibility, and plays a pivotal role in generating detailed test reports, providing valuable insights into the system's behavior and performance.

f. Simulation Engineer: Primary responsibility is to conduct Signal Integrity (SI), Power Integrity (PI), and thermal simulations. This involves a detailed analysis of simulation reports to assess the integrity of signals and power distribution within the system. The Simulation Engineer plays a crucial role in providing valuable insights for design improvements, collaborating closely with both hardware and firmware teams to optimize overall system performance.

g. Production Engineer: The key responsibilities revolve around efficient management of component procurement and the critical task of freezing components for the project. This involves close collaboration with suppliers to select suitable components and ensure their availability. Effective coordination with the project manager for necessary approvals is also a crucial aspect of this role, ensuring seamless progress and adherence to project goals.

h. Quality Assurance (QA) Engineer: The primary responsibilities encompass the meticulous verification of inspection reports and functional test reports. This role extends to conducting regular audits throughout the embedded project development process, ensuring adherence to quality standards and compliance regulations. The QA Engineer takes a proactive stance by implementing corrective and preventive actions to address any identified issues. Furthermore, this role involves monitoring and effectively communicating quality metrics to the project team, fostering a continuous improvement mindset within the embedded project lifecycle.

i. Security Engineer: The core responsibilities involve the meticulous implementation of security measures for both hardware and firmware components. This includes ensuring secure boot verification and partitioning to safeguard critical elements of the system. The Security Engineer plays a pivotal role in certificate validation and the implementation of tamper protection mechanisms.

6 RACI

Process	Responsible	Accountable	Consulted	Informed
Requirement Analysis & Documentation	Hardware Engineer	Project Manager	Firmware Engineer, Simulation Engineer	Delivery Manager
High- & Low-Level Architecture Design	Hardware Engineer	Project Manager	Simulation Engineer	Delivery Manager
Schematic Design	Hardware Engineer	Project Manager	Firmware Engineer, Simulation Engineer, QA Engineer	Delivery Manager
Simulation Reports	Simulation Engineer	Project Manager	Hardware Engineer, Firmware Engineer, QA Engineer	Delivery Manager
Gerber Release Phase	Hardware Engineer	Project Manager	Simulation Engineer, Production Engineer	Delivery Manager
Hardware development	Test Engineer	Project Manager	Production Engineer	Delivery Manager
Firmware Development	Firmware Engineer	Project Manager	Hardware Engineer, Test Engineer	Delivery Manager

Mechanical Development	Mechanical Engineer	Project Manager	Hardware Engineer, QA Engineer	Delivery Manager
Pre compliance Test	Simulation Engineer	Project Manager	Hardware Engineer, Test Engineer	Delivery Manager
Beta Version Design	Hardware Engineer	Project Manager	Production Engineer, QA Engineer	Delivery Manager
Testing Phase	Test Engineer	Project Manager	Test Engineer, Firmware Engineer	Delivery Manager
UAT	Test Engineer	Project Manager	QA Engineer	Production Engineer

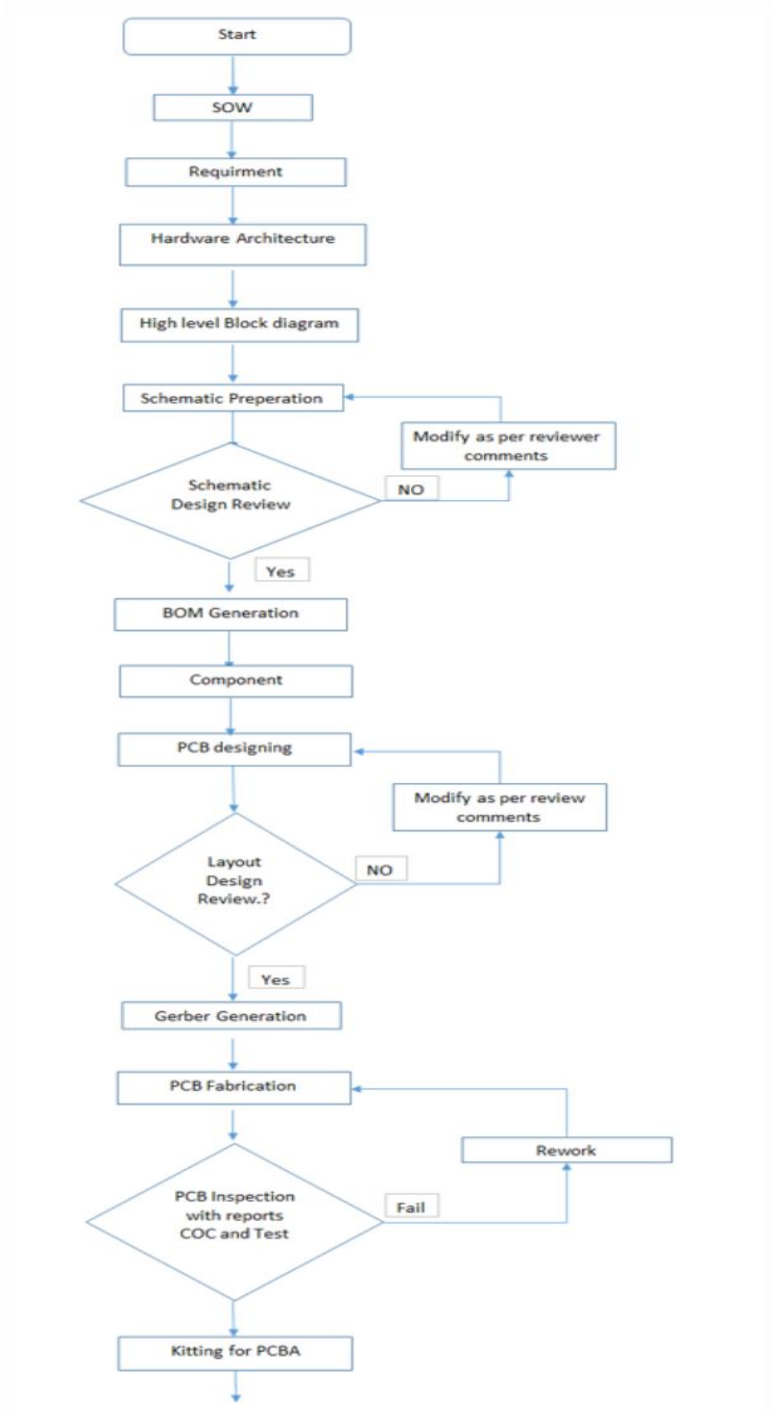
7 SIPOC

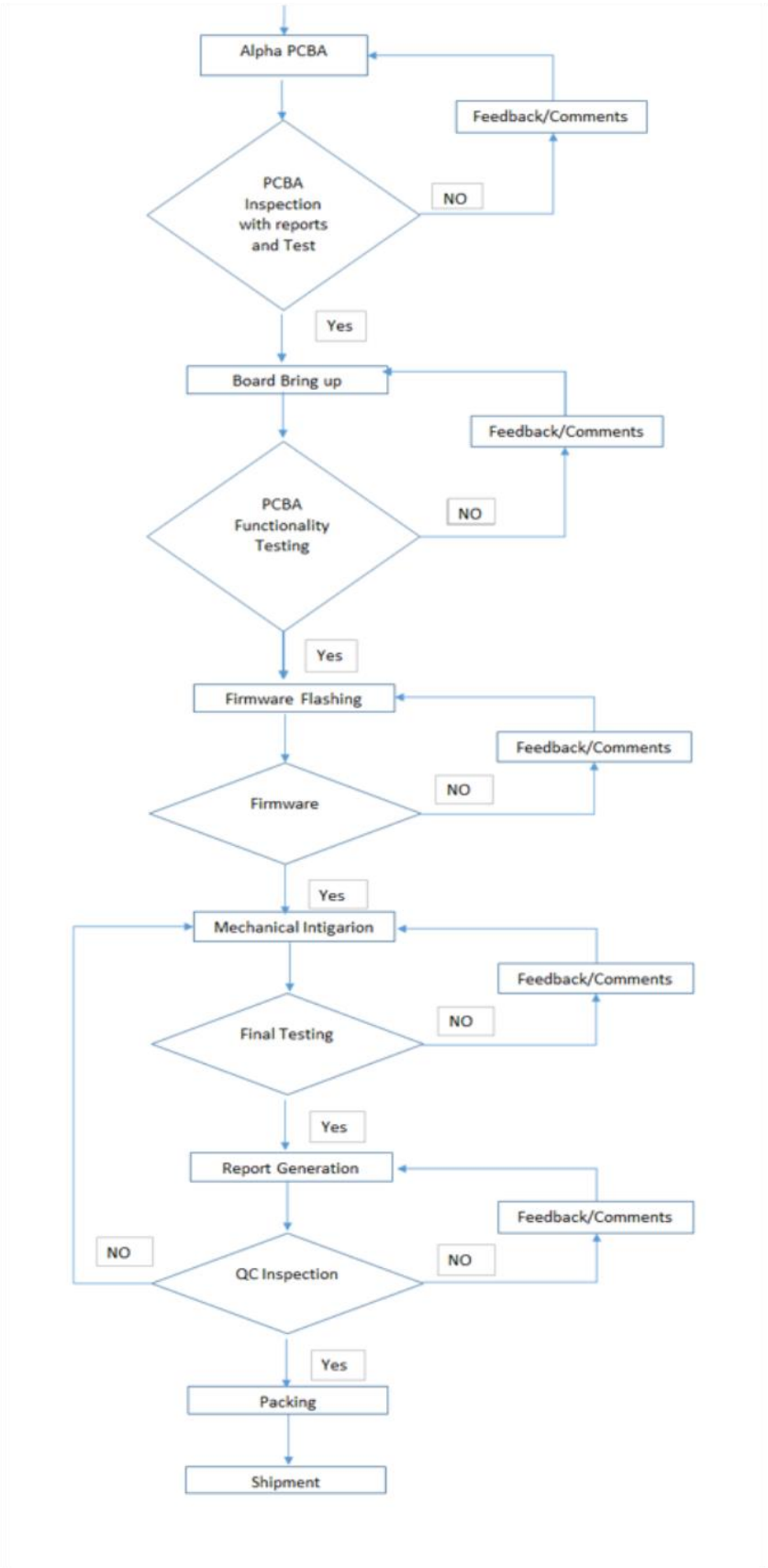
Supplier	Input	Process and Control		Output	Customer
		Process	Control		
Product owner	Requirements	Requirement Analysis & Documentation	Requirements	Requirement analysis & documentation documents	Project Manager, Firmware Engineer, Simulation Engineer, Delivery Manager
Hardware Engineer	Requirement Analysis & Documentation	High- & Low-Level Architecture Design	Architecture Design	Architecture design & documentation documents	Project Manager, Simulation Engineer, Delivery Manager

Hardware Engineer	Architecture Design	Schematic Design	Design analysis reports	Schematic & Design	Project Manager, Firmware Engineer, Simulation Engineer, QA Engineer,
Hardware Engineer	Schematic design	Simulation Reports	SI/PI report	Thermal Simulation	Project Manager, Firmware Engineer, QA Engineer,
Hardware Engineer	Schematic & Design analysis reports	Gerber Release Phase	Gerber release documents	Gerber Release	Project Manager, Simulation Engineer
Hardware Engineer	Completed Gerber release documents	Hardware development	Test cases	Hardware test reports	Project Manager, Firmware Engineer
Project Manager	Firmware requirements	Firmware Development	Firmware Requirement	Developed firmware	Project Manager, Hardware Eng
Project Manager	Mechanical Requirements	Mechanical Design	Mechanical requirements	Mechanical design documents	Project Manager, Hardware Engineer.
Project Manager	Hardware Test reports	Pre-Compliance Test	Test Reports	Pre-Compliance reports	Project Manager, Hardware Engineer, Test Engineer,
Project Manager	Pre-Compliance reports	Beta Version Design	Hardware Beta	Beta Reports	Project Manager, Hardware

					Engineer, Test Engineer,
Project Manager	Integrated Product	Testing Phase	Test Cases	Test Reports	Project Manager, Hardware Engineer, Firmware Engineer.
Project Manager	Integration Test Reports	UAT	Product Release	Released product	Project Manager, QA Engineer, Client/Vendor

8 Overall Process Flow Chart




Alpha PCBA
Feedback/Comments
PCBA Inspection with reports and Test
NO
Yes
Board Bring up
Feedback/Comments
PCBA Functionality Testing
NO
Yes
Firmware Flashing
Feedback/Comments
Firmware
NO
Yes
Mechanical Intigarion
Feedback/Comments
Final Testing
NO
Yes
Report Generation
Feedback/Comments
NO
QC Inspection
NO
Yes
Packing
Shipment

9 Process Description

9.1 Requirement Analysis & Documentation

a. The project team initiates the process by comprehensively understanding the scope of work and reviewing existing customer requirements, features, and functions.

b. A detailed examination is conducted by the team on any pre-existing designs or new requirements, deriving from both the existing product and potential enhancements, including exploration of novel features.

c. The project team shall formulate new hardware requirement specifications, considering aspects related to security and environmental considerations.

d. The team shall adhere to all the procedures outlined in Requirement Specification Template UTSDRSPTMP01.docx

e. Requirement specification will be reviewed using Requirement Specification Review Checklist UTSDRSPTMP02.xlsx and shall be approved by Architect / Product owner.

f. The finalized requirement specification shall be shared with the client for approval, along with the acceptance criteria for each requirement.

g. Upon client approval, the requirement specification is baselined and stored in the project repository.

9.2 High- & Low-Level Architecture Design

a. Project team will develop High-Level Architecture Design to define the overall architecture in detail, select components based on requirements, and assess feasibility.

b. Hardware lead or architect will detail the architecture blocks, outlining the main components and their interactions.

c. Design engineers shall select components, comparing at least two components per requirement, and apply Design Analysis Report, considering three different part numbers for each requirement.

d. The hardware lead shall execute a Proof of Concept (POC), if necessary.

e. The hardware engineer shall create a comprehensive hardware design document that includes detailed analysis of design calculations.

f. The proposed architecture and component selections shall undergo a thorough review by hardware design architect, ensuring alignment with the project's requirements and specifications.

g. The proposed architecture and component selections shall undergo a thorough review by hardware design architect, ensuring alignment with the project's requirements and specifications.

h. Project Team shall adhere to all the procedures outlined in Design Document Template UTSDDSPTMP01 ensuring that the documents undergo review and approval by Hardware design architect.

i. Design document will be reviewed using Design Review Checklist UTSDDSPTMP02.xlsx and shall be approved by Architect / Product owner.

j. The finalized design shall be shared with the client for approval, along with acceptance criteria for each requirement.

k. Upon client approval, the design is baselined and stored in the project repository.

l. The hardware design team shall initiate an in-depth analysis phase post the design approval.

m. The hardware team shall systematically conduct various assessments and generate reports to ascertain the resilience and reliability of the design.

n. The Project team shall perform a Failure Modes and Effects Analysis (FMEA) to identify and prioritize potential failure modes.

o. The Project team calculates the Mean Time Between Failures (MTBF) and implements derating to optimize component performance, with a crucial consideration for life cycle aspects.

p. The resulting reports shall be reviewed and approved by the client to ensure transparency and alignment with the project requirements.

9.3 Schematic Design

a. Design engineer shall Identify key components and their interconnections based on the project scope.

b. Design engineers shall choose an appropriate schematic capture tool (e.g., Altium, OrCAD) based on project requirements and team expertise, to ensure compatibility with other tools in the hardware development workflow.

c. Engineers shall use electronic components based on the identified functional blocks and project specifications, considering factors such as component availability, cost, and compatibility with the overall system.

d. Design engineers shall create symbols for selected components within the chosen schematic capture tool's library, to ensure that symbols accurately represent the electrical characteristics of the components.

e. Design engineers shall develop the schematic diagram by placing and connecting symbols on the schematic canvas, to follow a clear and organized layout to represent the logical flow of the circuit.

f. Engineers should perform Design Rule Checks to identify and resolve any errors or violations in the schematic, to ensure compliance with design guidelines, electrical constraints, and industry standards.

g. The hardware lead should conduct an internal review of the schematic with the hardware design team, to verify that the schematic meets the project requirements, and the team will address any identified issues or improvements.

h. The hardware manager will share the schematic with the client for review and feedback.

i. The Hardware Engineer shall gather input on specific design choices and ensure alignment with the client's expectations.

j. Upon client approval, the team will generate preliminary 3D models of the components to assess physical placement, considering spatial constraints and ensure that components fit within the intended enclosure.

k. Design engineers shall document the schematic design, including component values, reference designators, and any design notes.

l. Project manager should share the final schematic design with the client for a final round of approval, the team will proceed with PCB layout post customer Signoff.

m. The hardware team should verify the list of components identified and approved during the design phase to confirm that the specifications of the components match the design requirements.

n. Production engineer should prepare request for quotation (RFQ) document detailing the required components, quantities, and specifications and send the RFQ to potential suppliers, including relevant project details and delivery expectations.

o. Production Engineers evaluate responses from suppliers based on factors such as cost, lead time, reputation, quality and select suppliers that best meet project requirements and quality standards.

p. Purchase managers will negotiate terms and conditions, including pricing, delivery schedules, and payment terms and reach agreements with selected vendors from registered vendors list and document the terms in a formal contract or purchase order.

q. Purchase managers generate and issue purchase orders to selected suppliers, specifying the components, quantities, agreed-upon prices, and delivery dates.

r. Purchase managers monitor the status of orders regularly to track procurement progress and communicate with suppliers to address any issues and ensure timely delivery.

s. Admin team will receive components and hand them over to the lab team as they arrive from suppliers.

t. Production engineers will conduct thorough IGI inspections to verify that the received components match the specified requirements and are free from defects.

u. Lab team will update the inventory system with the received components and Maintain tracker for manage inventory levels to ensure availability for production.

v. The quality assurance team will verify that processes, procedures, and activities are in place to guarantee that products or services adhere to established quality standards and requirements.

w. The hardware team utilizes the finalized design specifications and analysis outcomes to conclude the PCB design and schematic.

x. Team develops a comprehensive PCB floor plan, outlining the placement of major components and their interconnections.

y. Hardware engineers place components on the PCB layout, ensuring proper positioning and connectivity and conduct design rule checks (DRC) to validate adherence to design guidelines.

z. Team generates preliminary 3D files to assess the feasibility of the physical layout and component placements.

aa. Team Present the preliminary design to the client for review, incorporating any feedback or adjustments, as necessary.

bb. Team documents the guidelines and specifications for the PCB layout, ensuring consistency and adherence to standards, and establish trace routes on the PCB, ensuring proper length matching for signal integrity and performance. Hardware lead confirms the finalization of schematic design files, incorporating any changes or optimizations.

cc. Hardware managers conduct a thorough review of the entire hardware development process with the client, addressing any concerns or modifications and requesting formal approval before proceeding with Gerbers.

9.4 Simulation Reports

a. Engineer performs a thorough Signal Integrity (SI) analysis to evaluate the quality of electrical signals within the system. Identify potential issues related to signal distortion, noise, or interference.

b. Simulation Engineer executes Power Integrity (PI) analysis to ensure stable power distribution throughout the system. Evaluate factors such as voltage droops, power noise, and the impact on the overall functionality.

c. Engineers conduct thermal simulations to assess the temperature distribution within the hardware components.

d. Identify potential hotspots and areas where cooling is required.

e. Simulation Engineer summarizes the findings from the SI, PI, and Thermal simulations in comprehensive reports. These reports include detailed analyses, graphical representations, and recommendations for improvements.

9.5 Gerber Release Phase

a. The hardware development team finalizes the Alpha PCB layout, ensuring that it aligns with the approved schematic design and meets all design requirements.

b. Team generates Gerber files, which include the necessary information for PCB fabrication, including copper layers, solder mask, and silkscreen. Simultaneously, create assembly files that detail component placements and orientations.

c. Hardware engineer Produce additional supporting files that aid in the assembly process, such as centroid files, which provide the coordinates for component placement. These files are crucial for accurate and efficient PCB assembly.

d. Hardware lead conduct an internal review of the finalized Gerber files, assembly files, and supporting documents to ensure accuracy and completeness. Once the internal review is complete, submit the documentation to the client for a thorough review and approval.

e. The client reviews the Gerber files, assembly files, and supporting documents, if any comments or modifications are addressed promptly. Upon client approval, the Gerber release phase was considered completed.

9.6 Hardware Development

a. Hardware engineer shall Initiate the fabrication process for the Alpha PCB based on the finalized Gerber files.

b. Hardware engineer shall Collaborate with the fabrication facility to ensure the accurate reproduction of the PCB layout.

c. The Hardware engineer shall assemble the fabricated PCB with the selected components.

d. ensure to Follow the Bill of Materials (BOM) and assembly guidelines to correct the component placement.

e. Hardware engineer shall Initiate the fabrication process for the Alpha PCB based on the finalized Gerber files.

f. Hardware engineer shall Collaborate with the fabrication facility to ensure the accurate reproduction of the PCB layout.

g. The Hardware engineer shall assemble the fabricated PCB with the selected components.

h. ensure to Follow the Bill of Materials (BOM) and assembly guidelines to correct the component placement.

i. The hardware engineer shall conduct the board bring-up process to verify the functionality of the assembled PCB. This involves powering up the board and checking for any initial issues or malfunctions.

j. Production engineers shall perform visual inspections of the assembled PCB to identify any visible defects or anomalies. Additionally, physical inspections will be done to ensure proper component soldering and overall board integrity.

k. The hardware testing team execute sanity testing to ensure basic functionalities are working as expected. Subsequently, comprehensive functional testing to validate the overall performance of the Alpha prototype.

l. Team executes performance testing for the prototype under various conditions, including thermal aspects. This involves evaluating the board's behavior under stress and ensuring it meets specified thermal requirements.

m. The hardware testing team will document the results of all testing phases in a comprehensive prototype test report. Including details on the tests conducted, observed outcomes, and any deviations from expected behavior.

n. The hardware lead conducts an internal review of the prototype test report to ensure accuracy and completeness. The report shall be submitted to the client for their thorough review and approval. the team will address any client feedback or concerns.

o. The hardware team shall conduct the security implementation process to enhance hardware security features. This involves addressing vulnerabilities and securing critical aspects of the hardware design.

p. Hardware Security Team shall Implement secure coding practices by signing and programming the code securely. cryptographic methods shall be used to ensure the integrity and authenticity of the code.

q. The Hardware Security Team verifies the secure boot process to ensure that the hardware system only boots using trusted and authenticated firmware. This step prevents unauthorized or malicious firmware from being loaded during the boot-up sequence.

r. Team shall Implement secure partitioning to isolate different components or functions within the hardware. This adds an additional layer of protection, preventing unauthorized access or interference.

s. Team shall Implement firmware rollover mechanisms to facilitate secure and seamless updates. This ensures that firmware updates can be performed securely, minimizing the risk of vulnerabilities.

t. The Hardware engineer Identifies and secures any unauthenticated parts of the application and shall apply security measures to protect sensitive data and functions from unauthorized access.

u. Hardware lead will Implement certificate validation processes to verify the authenticity and validity of certificates used in the hardware system. This helps in preventing the use of counterfeit or unauthorized certificates.

v. The hardware security engineer will enhance security by removing unnecessary services and libraries that may pose potential security risks. Streamlining the software components reduces the attack surface.

w. Hardware Engineer will Implement tamper protection mechanisms for interface devices to detect and respond to any physical tampering attempts. This adds an additional layer of security against unauthorized access.

x. The security engineer disables DMA (Direct Memory Access) to prevent unauthorized access to memory. This security measure restricts direct memory access and enhances the protection of sensitive data.

9.7 Firmware Development

a. Firmware team Initiate the process by documenting detailed firmware requirements based on project specifications and client needs.

b. Firmware engineer shall develop a high-level design document outlining the overall architecture, structure, and functionalities of the firmware.

c. The team Develop drivers for various peripherals and interfaces required for the functionality of the embedded system.

d. The firmware team Implement the main application logic based on the high-level and low-level design specifications.

e. The firmware engineer Integrate peripheral drivers with the application, ensuring proper communication and functionality and communication protocols and map variables/registers to facilitate data exchange.

f. The team shall Implement a bootloader for secure firmware updates, enabling a mechanism for updating firmware in deployed devices.

g. The team Integrate security measures, such as Trusted Platform Module (TPM) and digital certificates, to enhance the security of the firmware.

h. If hardware development is concurrent, the firmware engineer conducts board bring-up to ensure seamless integration with the hardware.

i. The firmware testing team conducts thorough unit testing to verify the functionality of individual components and modules.

j. The firmware testing team will address and resolve any identified bugs or issues through rigorous testing and debugging.

k. Firmware engineers will Perform code reviews to ensure adherence to coding standards and conduct verification and validation processes to confirm the firmware meets the specified requirements.

9.8 Mechanical development

a. The process shall begin with a comprehensive understanding of the requirements and the gathering of necessary information. This involves a detailed analysis of customer needs and project specifications.

b. Mechanical engineers shall engage in brainstorming sessions to generate ideas and optimize design dimensions. The goal is to freeze the conceptual outline design, ensuring alignment with project objectives and client requirements.

c. The concept developed is reviewed by the mechanical lead and approved by the mechanical manager.

d. Following the outline of conceptualization, the engineer will prepare a rough Bill of Materials (BOM).

e. The design phase shall involve establishing detailed design of concept specifications and preparing for the preliminary design. This step sets the foundation for the subsequent modeling and evaluation processes.

f. Engineers shall utilize the 3D modeling tools to create sub-assemblies and the main assembly.

g. The model undergoes thorough evaluation, considering parameters and criteria. Digital dimensions are derived from the model, and an internal review is conducted to assess the setup's viability and effectiveness.

h. This phase shall involve a meticulous evaluation of the conceptual model, with corrections made as needed. Specific considerations, such as providing holes for mounting should be addressed, and a comprehensive review of corrected models should take place. The conversion from 3D to 2D should be executed during this phase, providing a comprehensive design representation.

i. Final drawing preparation shall include a detailed review of 2D drawings, tolerance analysis, quality checks, and the release of drawings for manufacturing. This step ensures accuracy and completeness in the design documentation.

j. The team shall finalize the detailed BOM for ease of manufacturing. The ease of manufacturing shall be assessed for dimension and shape evaluations based on 2D drawings. Corrective measures shall be implemented for modifications required in the design and drawings. Assembling manufactured components follows, with overall dimensions and functionality checked.

k. The final steps shall involve improvisation based on testing outcomes, thorough validation to ensure adherence to specifications, and the formal handover of the validated design.

l. The design undergoes User Acceptance Testing (UAT), providing a final evaluation of the design's performance and functionality according to user requirements.

9.9 Pre-compliance test

a. Hardware team will prepare Hardware and required documents like User manual, Test application and test set up procedures for pre-compliance.

b. EMI/EMC test engineer shall identify the lab and get the Quotation for the required test and finalize the laboratory by checking the lab slot availability.

c. EMI/EMC test engineer initiates pre-compliance testing process by conducting Electromagnetic Interference (EMI) and Electromagnetic Compatibility (EMC) tests in a controlled laboratory environment. These tests are essential to assess the impact of the device on the electromagnetic spectrum and ensure compliance with relevant standards.

d. Team Perform EMI/EMC testing shall involve subjecting the Alpha Unit to various electromagnetic conditions to evaluate its performance in different scenarios. This includes testing for emissions and susceptibility to interference.

e. The team shall document comprehensive test reports detailing the observations, findings, and results of the EMI/EMC tests. Report shall Include information on any identified issues, deviations from standards, and areas of compliance.

f. The team shall conduct an internal review of the generated test reports. This review shall involve a thorough examination of the documented observations, ensuring accuracy and completeness of the information.

g. For failures, identify the root cause and fix the issues in the current revision by adding appropriate filters and rerun the test.

h. Project manager shall share the test reports with the client for their review and approval. The client's input is crucial to ensuring that the Alpha Unit meets the required EMI/EMC standards and specifications.

i. The team shall address any feedback or concerns provided by the client during the review process. The team shall make necessary adjustments or improvements to address identified issues and enhance the overall compliance of the Alpha Unit.

j. The team shall finalize the test reports based on the internal review, client feedback, and any necessary adjustments. Ensure that the documentation accurately reflects the EMI/EMC performance of the Alpha Unit.

k. Project manager shall submit the finalized test reports to the client for final approval. Client approval signifies acceptance of the EMI/EMC compliance of the Alpha Unit.

l. The team shall document the compliance status of the Alpha Unit based on the approved test reports. Clearly indicate whether the device meets the required EMI/EMC standards and specifications.

m. Project manager shall archive the approved test reports and related documentation in the project repository. The documentation shall serve as a reference for future compliance assessments and regulatory requirements.

9.10 Beta Version Design

a. Hardware engineer shall develop the beta version schematics based on the finalized design specifications on any updates or changes required for the beta release. Ensure that the schematics accurately represent the intended changes and improvements.

b. The hardware team shall prepare the Beta BOM, specifying all the components required for the beta version. Documents shall include details such as component names, quantities, and suppliers, facilitating the procurement process.

c. Team shall Implement the beta version PCB layout based on the updated schematics. Considering the changes and improvements introduced in the beta release, ensuring that the layout aligns with the design specifications.

d. The hardware design engineer shall generate the Gerber files for the beta version PCB layout. These files are essential for the PCB fabrication process and provide the necessary information for manufacturing the beta boards.

e. Lab manager shall Initiate the procurement process for the required components specified in the Beta BOM. Manager shall share the beta version PCB layout files to the fabrication facility for manufacturing.

f. The hardware team shall assemble the beta version PCBs by placing and soldering the components onto the fabricated boards. Team will Follow industry standards and best practices for PCB assembly to ensure the reliability of the beta units.

g. The hardware engineer shall conduct the board bring-up process to verify the functionality of the beta version. This involves powering up the boards, checking for any initial issues, and ensuring that the hardware responds as expected.

h. Hardware testing team shall Perform comprehensive functional testing on the beta version to validate that all hardware components and features operate according to the design specifications. Identify and address any functional issues or deviations and record the same.

i. The hardware testing team executes beta performance testing to assess the overall performance of the hardware under different conditions. This may involve testing for speed, efficiency, and any specific performance criteria defined for the beta release.

j. The testing team shall document the results of the design verification testing for the beta version. This document shall serve as evidence that the hardware meets the specified design requirements and has undergone thorough testing.

k. Team shall conduct pre-compliance testing on the beta version to evaluate its adherence to relevant standards before the final compliance testing. The team shall generate a pre-compliance test report outlining the results and any necessary corrective actions.

l. The hardware team shall document all layout changes introduced in the beta version, including updated schematics, Gerber files, and any modifications made during the design and testing phases.

m. Hardware manager shall Review the beta version design and layout change document with relevant stakeholders, including the client. The team shall get approval for the changes made and ensure alignment with project requirements.

9.11 Testing Phase

a. Test Engineer shall Develop detailed test cases based on the functional specifications and design documentation. Each test case should cover specific functionalities and potential edge cases.

b. Test lead shall Establish the test environment, including the necessary hardware, software, and any testing tools. Ensure that the environment replicates the conditions under which the embedded system will operate.

c. Test engineer shall Execute the prepared test cases, systematically validating the functionality of the embedded system. Record the results, including any deviations or failures encountered during testing.

d. Defect Reporting: Document and report any identified defects or issues. Provide detailed information on the nature of the defect, steps to reproduce, and potential impact on the system's functionality.

e. Validate the interactions and interoperability of different hardware and software components within the embedded system.

f. Integration Test Planning: Plan the integration testing strategy, identifying the components to be integrated and the order of integration. Define the criteria for successful integration.

g. Component Integration: Integrate individual hardware and software components step by step, following the predetermined plan. Verify that each integration step adheres to the specified interfaces and protocols.

h. Interface Testing: Specifically focus on testing the interfaces between components, ensuring proper communication and data exchange. Detect and address any inconsistencies or compatibility issues.

i. System Integration: Combine all integrated components to form the complete embedded system. Conduct comprehensive testing to validate the overall functionality and coherence of the system.

j. Regression Testing: Execute regression tests to confirm that new integrations have not adversely affected previously tested functionalities. Address any regression defects that arise during testing.

k. Test Lead shall Confirm that the embedded system meets the defined requirements and is ready for deployment.

l. Test Engineer shall Perform validation tests to ensure that the embedded system satisfies the specified requirements, including performance, reliability, and security criteria.

m. Project Manager or Test lead shall Review and update all relevant documentation, including test plans, test cases, and test results. Ensure that the documentation accurately reflects the final state of the embedded system.

n. Project Manager shall share the results of the final testing phase to the client for approval. comprehensive test reports, highlighting the testing process, outcomes, and any identified issues or resolutions shall be included as a part of the report.

o. Upon client approval, the Project Manager shall release the finalized embedded system for production or deployment. Provide all necessary documentation to support the ongoing maintenance and operation of the system.

9.12 UAT Support

a. PM shall maintain regular communication channels with the client to provide status updates on UAT progress.

b. Establish a systematic process for the client to report any issues or defects identified during UAT.

c. Report any issues, challenges, or delays promptly and work together to find solutions.

d. Log and prioritize reported issues and work collaboratively with the development and testing teams to address and resolve them.

e. Document the UAT results, including successful test cases, identified issues, and resolutions.

f. Provide knowledge transfer sessions to the client's team for a smooth transition to the production phase.

g. Gather feedback from the client regarding the UAT process and overall project collaboration.

h. Conduct a retrospective analysis to identify areas for improvement and incorporate lessons learned into future projects.

10 Information Security requirements

a. The information security related requirements shall be included in the Product backlog for added information systems or enhancements to existing information systems.

b. Secure access (user registration and deregistration) to various applications, Project portals

c. Information Security requirements in the contract are clearly put into place.

d. Compliance requirements based on different geographies are adhered to in projects.

e. Confidentiality of information and project details is maintained by the team.

11 References

a. SOP Requirement Specification

b. SOP Design

c. SOP Coding

d. SOP Review Process

e. SOP Testing Validation and Acceptance

12 Metrics

a. Build success rates, Deployment frequency, Release cycles.

b. Development productivity metrics

c. Code coverage

d. Test pass rate

e. Defect density

13 Annexures

a. Nil

14 Forms

a. Embedded review checklist UTSDHPTMP03

b. Review Form UTPSRVPTMP02

c. Requirement Specification Template UTSDRSPTMP01

d. Requirement Specification Review Checklist UTSDRSPTMP02

e. Design Document Template UTSDDSPTMP01

f. Design Review Checklist UTSDDSPTMP02.